

6 a (ii)1 $E = \frac{W}{Q} = \frac{2.4 \times 10^5}{2.2 \times 10^4} = 11 \text{ V}$ M1

2 $Q = It = \frac{E}{R} t$
 $\Rightarrow t = \frac{QR}{E} = \frac{2.2 \times 10^4 \times 3800}{11}$ M1

$= 7.6 \times 10^6 \text{ s}$ A1

3 fraction = $\frac{I^2 R_T}{I^2 R_T + I^2 R} = \frac{R_T}{R_T + R} = \frac{1800}{1800 + 2000}$ M1
 $= 0.474$ A1

Note: fraction must be written in decimal. -1 for answer not written in decimal

b (i) $V_{XY} = \frac{8.0}{8.0 + 0.50 + 2.5} \times 2.0 = 1.455 \text{ V}$ M1

At null deflection, V_{XJ} = e.m.f. of cell B.

$E = \frac{0.90}{1.0} \times 1.455 = 1.31 \text{ V}$ M1

(ii)1 At null deflection, $V_{XJ} = V_{RS}$ M1
 $V_{XJ} = \frac{0.75}{1.0} \times 1.455 = 1.091 \text{ V}$

$I_{RS} = \frac{V}{R} = \frac{1.091}{6.5} = 0.1679 \text{ A}$ M1

$E = V + Ir \Rightarrow r = \frac{E - V}{I} = \frac{1.31 - 1.091}{0.1679} = 1.30 \text{ } \Omega$ A1

allowed range of E: 1.20 to 1.40 depending on substitution

With resistor connected in parallel, load resistance of cell B would decrease.

B1

This would cause terminal p.d. of cell B to decrease.

Since no change in potential difference per unit length along wire XY,

A1

Balance length XJ would decrease.

